Machine Learning - Assignment 2

Dry Bean Classification - Submission

M.Tech (AIML / DSE), Work Integrated Learning Programmes Division

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1. GitHub Repository Link

The complete source code, requirements.txt, README.md, and test data (CSV) are hosted at:

https://github.com/2025da04290/ml-assignment-2-dry-beans

Repository contents:
- app.py (Streamlit web application)
- train_models.py (training script - generates models + test_data.csv)
- requirements.txt (Python dependencies)
- README.md (problem statement, dataset description, comparison table, observations)
- test_data.csv (held-out test set used in the Streamlit app)
- model/ (saved .joblib files for all 5 models, scaler, label encoder, metrics_summary.csv, and the training notebook model_training.ipynb)

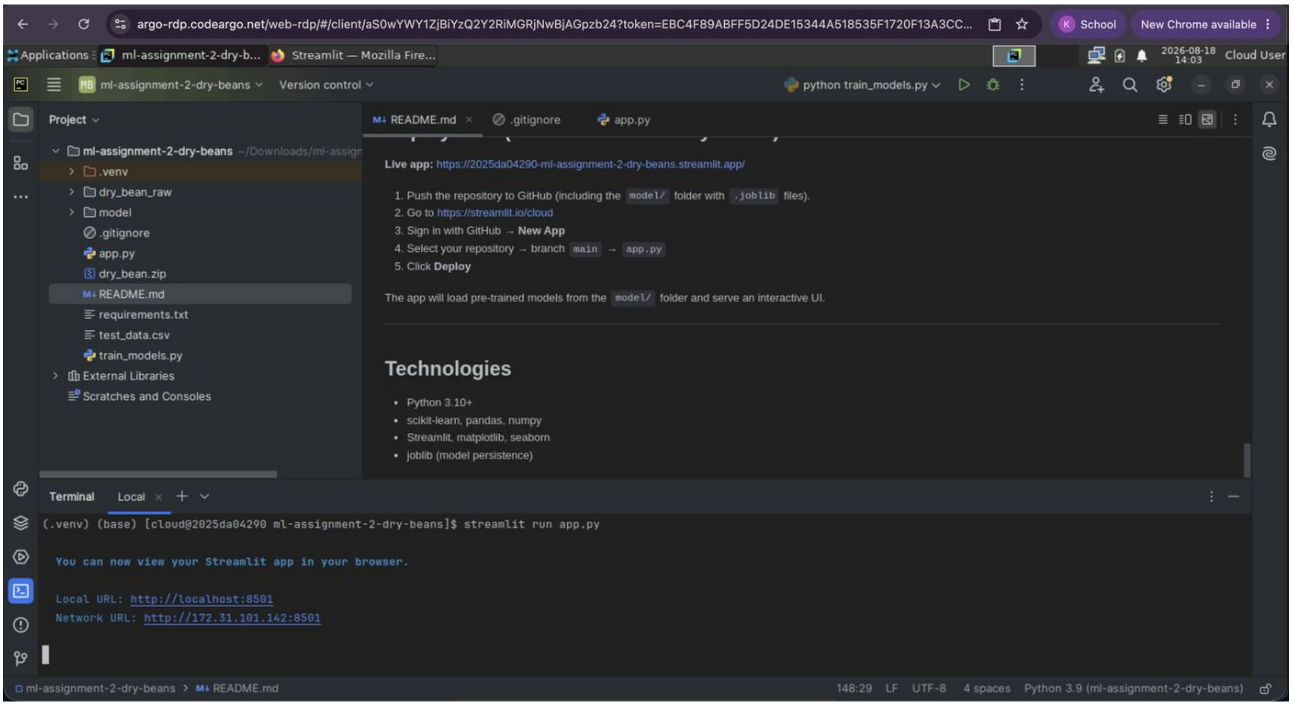
2. Live Streamlit App Link

Deployed using Streamlit Community Cloud. Opens an interactive frontend when clicked.

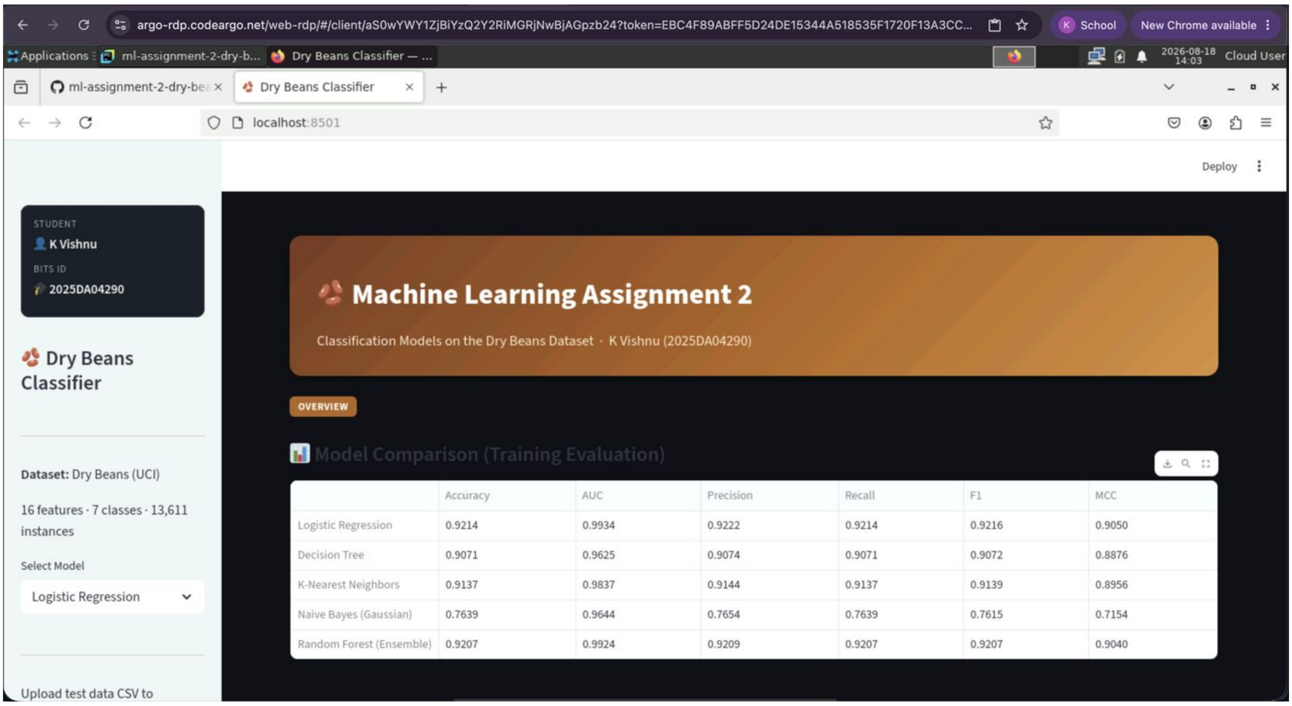
https://2025da04290-ml-assignment-2-dry-beans.streamlit.app/

3. Screenshot

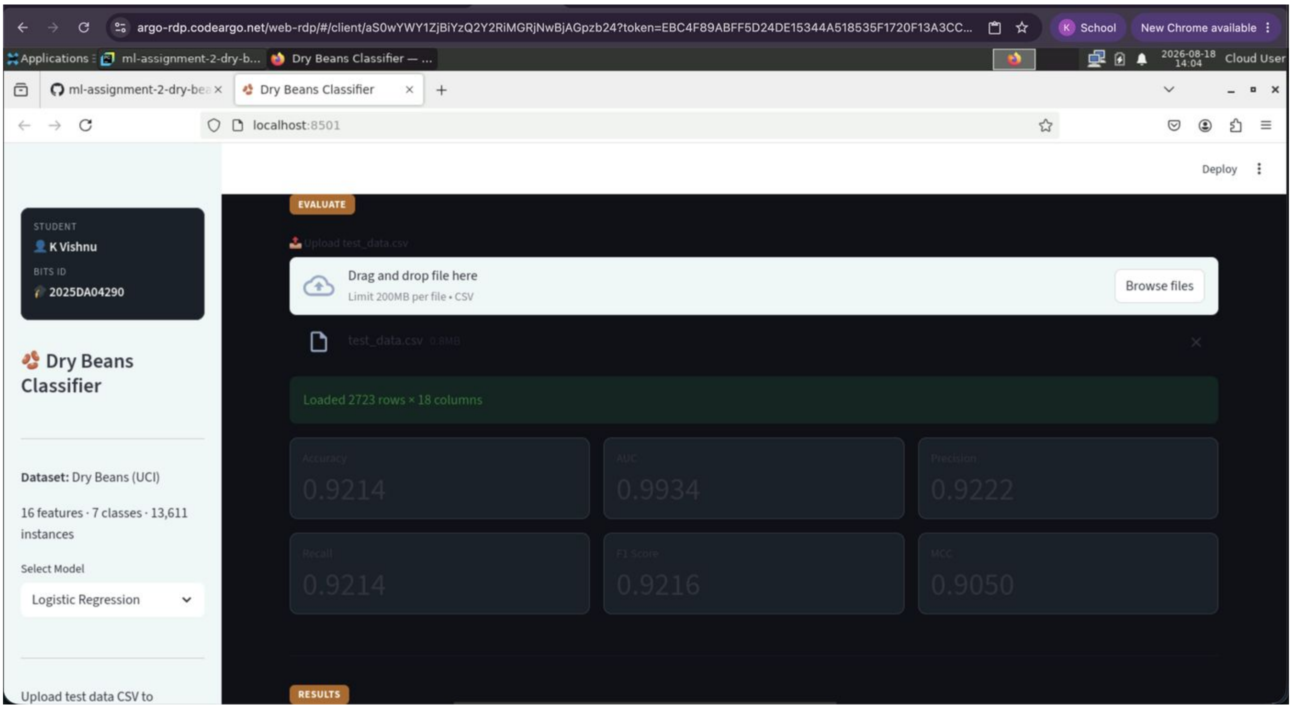
Screenshots of assignment execution on the BITS Virtual Lab (RDP session), captured end-to-end from launching the app through to verifying the pushed GitHub repository.



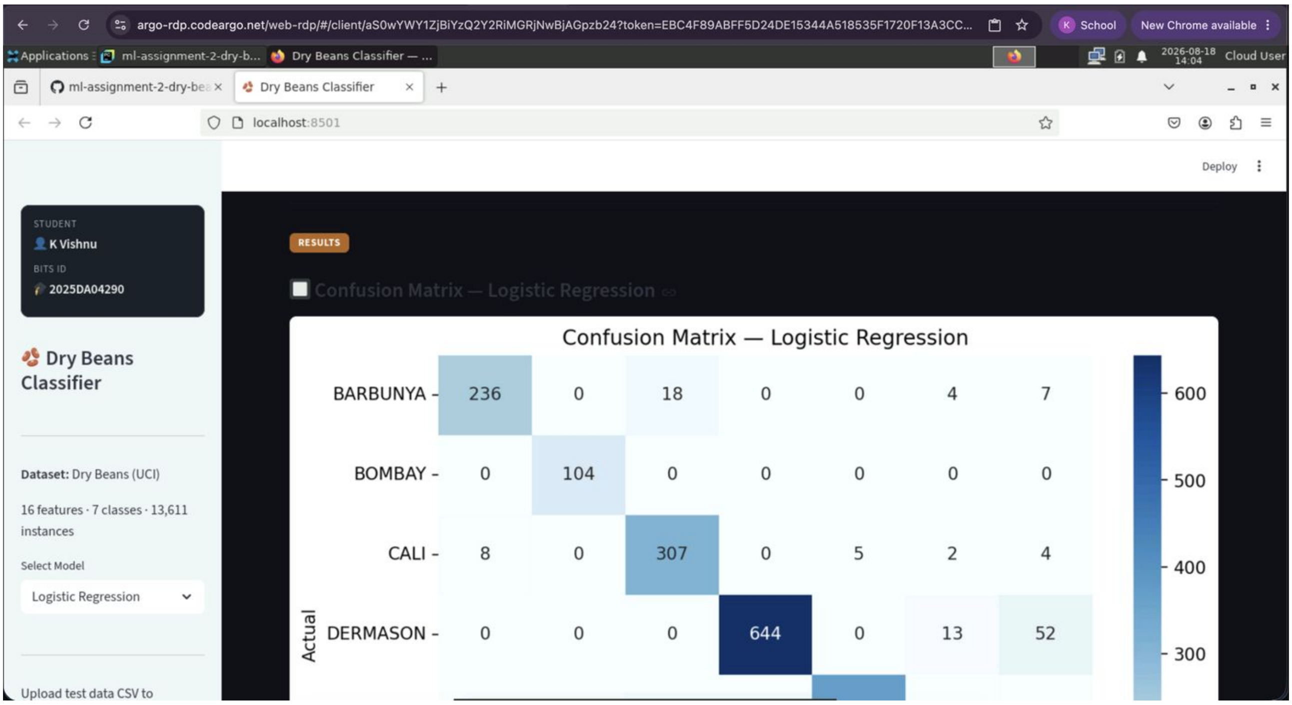
Step 1. *BITS Virtual Lab (RDP session): PyCharm project open, running streamlit run app.py in the integrated terminal.*



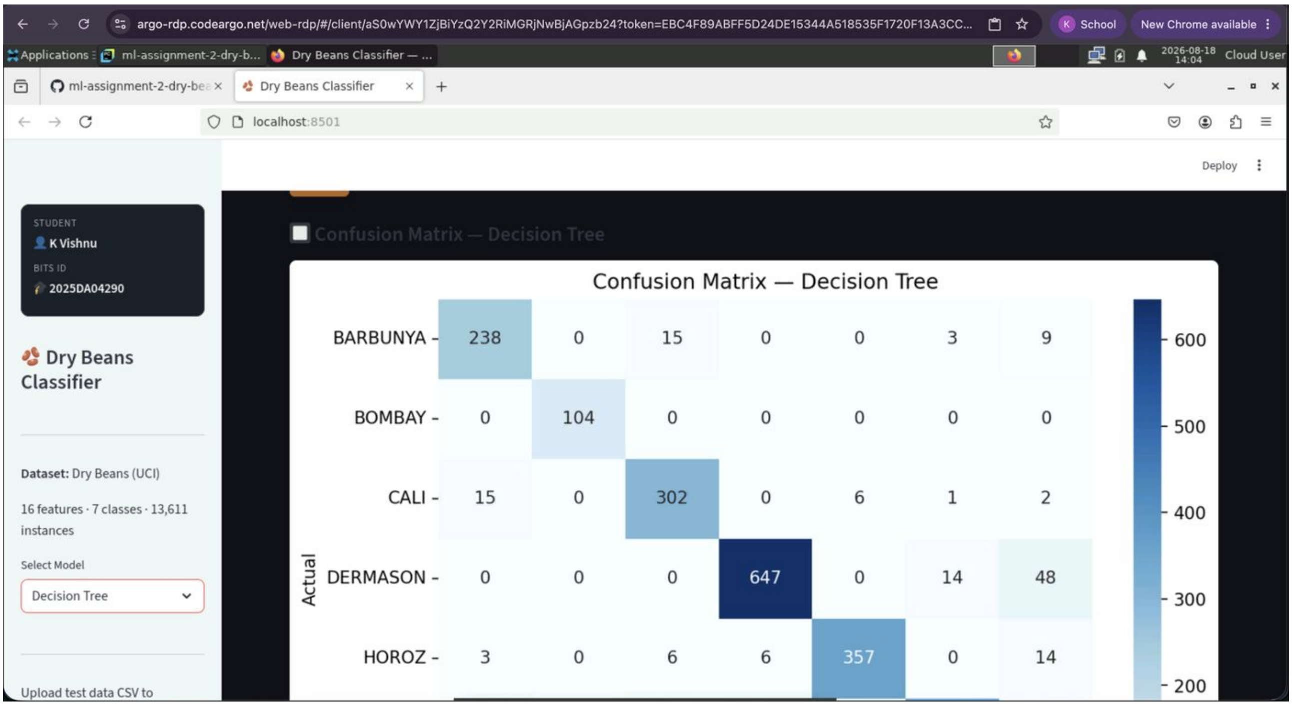
Step 2. *App running at localhost:8501 inside the lab session: student info card, hero header, and the 5-model comparison table.*



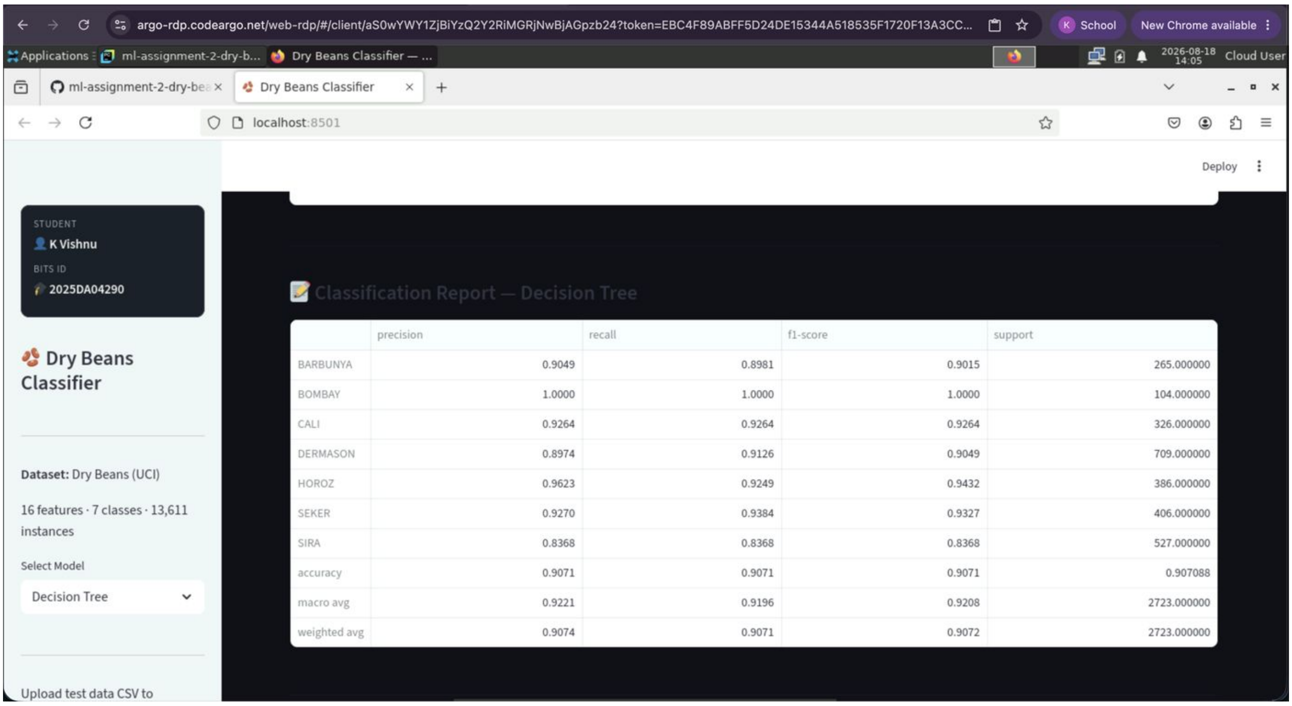
Step 3. *test_data.csv uploaded (2,723 rows x 18 columns) and evaluation metrics computed for Logistic Regression.*



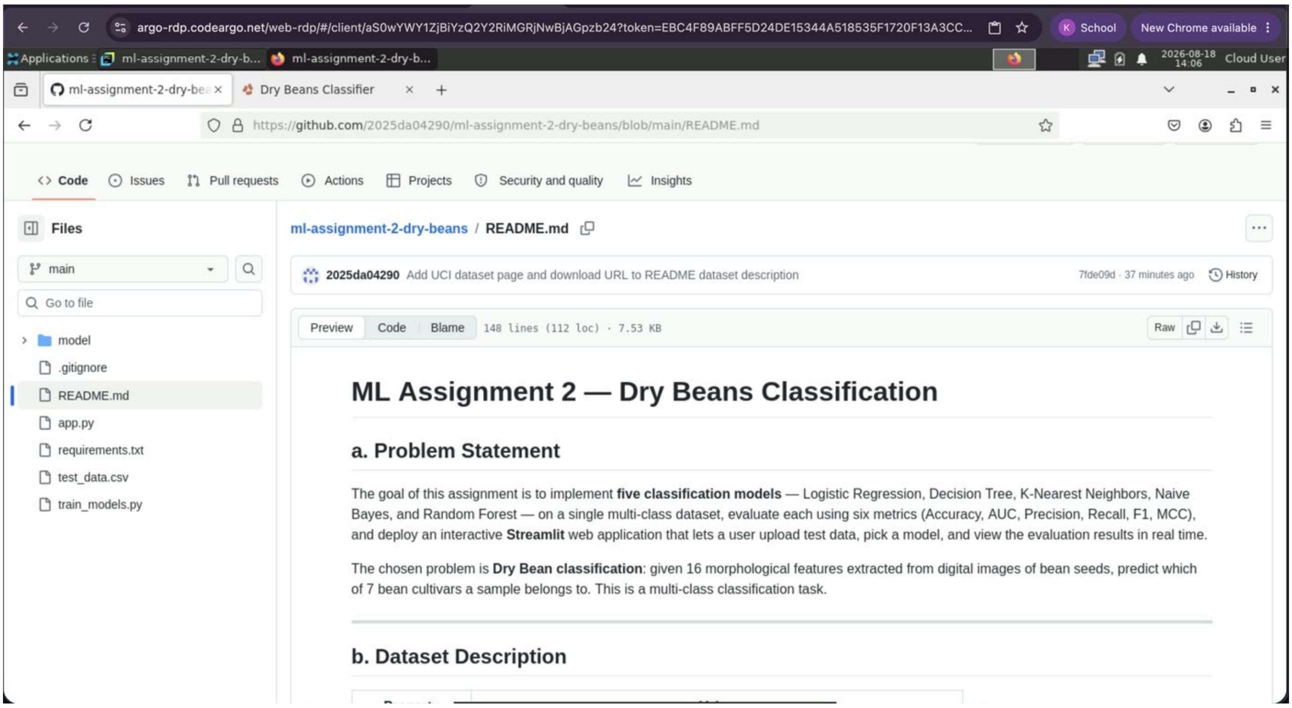
Step 4. *Confusion matrix rendered for Logistic Regression.*



Step 5. *Model switched to Decision Tree via the sidebar dropdown; confusion matrix updates accordingly.*



Step 6. *Classification report (precision/recall/F1 per class) for Decision Tree.*



Step 7. *GitHub repository (README.md) viewed from within the BITS Lab session, confirming the pushed repository.*

4. GitHub README Content

a. Problem Statement

The goal of this assignment is to implement five classification models - Logistic Regression, Decision Tree, K-Nearest Neighbors, Naive Bayes, and Random Forest - on a single multi-class dataset, evaluate each using six metrics (Accuracy, AUC, Precision, Recall, F1, MCC), and deploy an interactive Streamlit web application that lets a user upload test data, pick a model, and view the evaluation results in real time.

The chosen problem is Dry Bean classification: given 16 morphological features extracted from digital images of bean seeds, predict which of 7 bean cultivars a sample belongs to. This is a multi-class classification task.

b. Dataset Description

Name	Dry Beans
Source	UCI Machine Learning Repository (ID: 602)
Dataset page	archive.ics.uci.edu/dataset/602/dry+bean+dataset
Download URL	https://archive.ics.uci.edu/static/public/602/dry+bean+dataset.zip
Instances	13,611
Features	16 (all numeric)
Classes	7 (Barbunya, Bombay, Cali, Dermason, Horoz, Seker, Sira)
Task Type	Multi-class classification

Feature list (16 numeric features):

#	Feature	Description
1	Area	Area of a bean zone (pixels)
2	Perimeter	Bean perimeter (pixels)
3	MajorAxisLength	Length of major axis
4	MinorAxisLength	Length of minor axis
5	AspectRatio	Major/minor axis ratio
6	Eccentricity	Eccentricity of the ellipse
7	ConvexArea	Convex hull area
8	EquivDiameter	Equivalent diameter
9	Extent	Ratio of pixels in bounding box
10	Solidity	Ratio of area to convex area
11	roundness	Roundness measure
12	Compactness	Compactness measure
13	ShapeFactor1	Shape factor 1
14	ShapeFactor2	Shape factor 2
15	ShapeFactor3	Shape factor 3
16	ShapeFactor4	Shape factor 4

The dataset satisfies the assignment constraints: 16 features >= 12 minimum, 13,611 instances >= 500 minimum.

c. GitHub Repository Link

https://github.com/2025da04290/ml-assignment-2-dry-beans

d. Models Used - Comparison Table

ML Model	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9214	0.9934	0.9222	0.9214	0.9216	0.9050
Decision Tree	0.9056	0.9619	0.9060	0.9056	0.9057	0.8858
K-Nearest Neighbors	0.9137	0.9837	0.9144	0.9137	0.9139	0.8956
Naive Bayes (Gaussian)	0.7639	0.9644	0.7654	0.7639	0.7615	0.7154
Random Forest (Ensemble)	0.9203	0.9924	0.9205	0.9203	0.9203	0.9036

Observations on Model Performance

ML Model	Observation about model performance
Logistic Regression	The top performer on Accuracy/Precision/Recall/F1/MCC (~92.1% accuracy, MCC 0.9050) and highest AUC (0.9934) of all five models. The linear decision boundaries work well because many bean morphological features are roughly linearly separable after scaling.
Decision Tree	Weakest of the tree/linear/instance-based group at ~90.6% accuracy and the lowest AUC among the non-Naive-Bayes models (0.9619), since a single tree gives coarse, discrete leaf probabilities and is prone to overfitting on majority classes (Dermason, Sira).
K-Nearest Neighbors	Solid performance at ~91.4% accuracy with k=7 on scaled features - the scaling step is essential, since raw-feature distances would be dominated by high-magnitude features like Area. Struggles most on overlapping classes (Sira vs Dermason).
Naive Bayes (Gaussian)	The clear weakest performer (~76.4% accuracy, MCC 0.7154), despite a respectable AUC (0.9644) - the Gaussian conditional-independence assumption is badly violated here since many features (Area, ConvexArea, Perimeter, EquivDiameter) are highly correlated, which hurts its hard-label predictions much more than its probability ranking.
Random Forest (Ensemble)	Very close second overall (~92.0% accuracy, MCC 0.9036), narrowly behind Logistic Regression on the threshold-based metrics but essentially tied on AUC (0.9924 vs 0.9934). The ensemble of 200 trees captures non-linear feature interactions and reduces the variance of the single Decision Tree substantially.
Overall Winner for your dataset?	Logistic Regression - it edges out Random Forest on every metric except being effectively tied on AUC. This reflects that the 16 shape/size features are largely linearly separable across the 7 bean classes once standardized, so the extra model capacity of an ensemble of trees isn't needed here. Random Forest remains a very close second and the more robust choice if the feature-class relationship were less linear.